1. Запуск landing_to_bronze.py

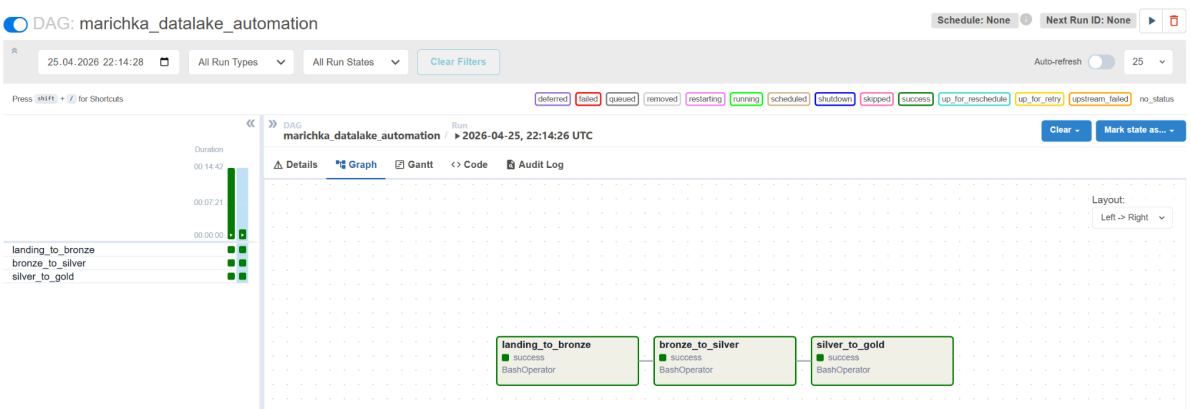
```
 applicable
Завантаження https://ftp.goit.study/neoversity/athlete_bio.csv...
Збережено в data/landing/athlete_bio.csv
Обробка таблиці athlete_bio...
26/04/25 21:40:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 95.00% for 8 writers
26/04/25 21:40:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 84.44% for 9 writers
26/04/25 21:40:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 76.00% for 10 writers
26/04/25 21:40:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 69.09% for 11 writers
26/04/25 21:40:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 63.33% for 12 writers
26/04/25 21:40:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 58.46% for 13 writers
26/04/25 21:40:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 54.29% for 14 writers
26/04/25 21:40:46 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 58.46% for 13 writers
26/04/25 21:40:46 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 63.33% for 12 writers
26/04/25 21:40:46 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 69.09% for 11 writers
26/04/25 21:40:46 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 76.00% for 10 writers
26/04/25 21:40:46 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 84.44% for 9 writers
26/04/25 21:40:46 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 95.00% for 8 writers
Таблицю athlete_bio успішно збережено в Bronze (Parquet) за шляхом: data/bronze/athlete_bio
Завантаження https://ftp.goit.study/neoversity/athlete_event_results.csv...
Збережено в data/landing/athlete_event_results.csv
Обробка таблиці athlete_event_results...
26/04/25 21:41:03 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 95.00% for 8 writers
Таблицю athlete_event_results успішно збережено в Bronze (Parquet) за шляхом: data/bronze/athlete_event_results
(airflow_env) marichka@DESKTOP-SFONK0I:~/datalake_project$
```

2. Запуск `bronze_to_silver`

```
26/04/25 21:43:37 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-j
 applicable
Обробка athlete_bio: Bronze -> Silver...
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 95.00% for 8 writers
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 84.44% for 9 writers
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 76.00% for 10 writers
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 69.09% for 11 writers
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 63.33% for 12 writers
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 58.46% for 13 writers
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 54.29% for 14 writers
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 50.67% for 15 writers
26/04/25 21:43:43 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 47.50% for 16 writers
26/04/25 21:43:44 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 50.67% for 15 writers
26/04/25 21:43:44 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 54.29% for 14 writers
26/04/25 21:43:44 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 58.46% for 13 writers
26/04/25 21:43:44 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 63.33% for 12 writers
26/04/25 21:43:44 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 69.09% for 11 writers
26/04/25 21:43:44 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 76.00% for 10 writers
26/04/25 21:43:44 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 84.44% for 9 writers
26/04/25 21:43:44 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 95.00% for 8 writers
Таблицю athlete_bio очищено та збережено в Silver: data/silver/athlete_bio
```

```
Обробка athlete_event_results: Bronze -> Silver...
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 95.00% for 8 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 84.44% for 9 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 76.00% for 10 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 69.09% for 11 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 63.33% for 12 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 58.46% for 13 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 54.29% for 14 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 50.67% for 15 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 47.50% for 16 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 50.67% for 15 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 54.29% for 14 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 58.46% for 13 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 63.33% for 12 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 69.09% for 11 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 76.00% for 10 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 84.44% for 9 writers
26/04/25 21:43:45 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 95.00% for 8 writers
Таблицю athlete_event_results очищено та збережено в Silver: data/silver/athlete_event_results
```

3. Запуск **silver_to_gold**

```
(airflow_env) marichka@DESKTOP-SFONK0I:~/datalake_project$ python silver_to_gold.py
26/04/25 21:50:40 WARN Utils: Your hostname, DESKTOP-SFONK0I resolves to a loopback address: 127.0.1.1; using 10.255.255.254 instea
d (on interface lo)
26/04/25 21:50:40 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/25 21:50:41 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where
 applicable
Фінальну аналітичну таблицю успішно збережено в Gold: data/gold/avg_stats
+------------------+------+------+-----------+----------------+-----------------+--------------------+
|            sport| medal|   sex|country_noc|       avg_weight|        avg_height|           timestamp|
+------------------+------+------+-----------+----------------+-----------------+--------------------+
|Equestrian Jumping|  Gold|  Male|        USA|71.66666666666667|182.11111111111111|2026-04-25 21:50:...|
|        Ice Hockey|Bronze|  Male|        CZE|           89.84|           185.28|2026-04-25 21:50:...|
|           Archery|Silver|  Male|        FRA|            83.0|            180.0|2026-04-25 21:50:...|
|  Art Competitions|Bronze|  Male|        GER|            NULL|             NULL|2026-04-25 21:50:...|
|    Figure Skating|Bronze|  Male|        HUN|            63.0|            168.0|2026-04-25 21:50:...|
|          Shooting|Silver|  Male|        FRA|            75.8|            179.6|2026-04-25 21:50:...|
|           Sailing|  NULL|  Male|        ITA|78.32432432432432|179.26351351351352|2026-04-25 21:50:...|
|         Athletics|  NULL|  Male|        DOM|68.03174603174604| 174.0952380952381|2026-04-25 21:50:...|
|              Judo|  NULL|  Male|        USA|84.79310344827586|179.43103448275863|2026-04-25 21:50:...|
|      Canoe Slalom|  NULL|Female|        CZE|            59.0|165.4444444444444446|2026-04-25 21:50:...|
+------------------+------+------+-----------+----------------+-----------------+--------------------+
only showing top 10 rows
```

4. Запуск **project_solution через airflow**